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Studierichting: Informatica

Oefeningenreeks 1D nr. 23.8

$$A(z) = z^5 + (3 - 2i)z^4 + 2(1 - 3i)z^3 - 2(1 + 3i)z^2 - (3 + 2i)z - 1$$

$$A(-1) = -1 + 3 - 2i - 2 + 6i - 2 - 6i + 3 + 2i - 1 = 0 \Rightarrow z_1 = -1$$

$$\begin{array}{c|cccccc} & 1 & 3-2i & 2-6i & -2-6i & -3-2i & -1 \\ -1 & \downarrow & -1 & -2+2i & 4i & 2+2i & 1 \\ \hline & 1 & 2-2i & -4i & -2-2i & -1 & 0 \end{array}$$

$$A(z) = (z + 1)B(z) \rightarrow B(z) = z^4 + (2 - 2i)z^3 - 4iz^2 + (-2 - 2i)z - 1$$

$$B(-1) = 1 - 2 + 2i - 4i + 2 + 2i - 1 = 0 \Rightarrow z_2 = -1$$

$$\begin{array}{c|cccccc} & 1 & 2-2i & -4i & -2-2i & -1 \\ -1 & \downarrow & -1 & -1+2i & 1+2i & 1 \\ \hline & 1 & 1-2i & -1-2i & -1 & 0 \end{array}$$

$$B(z) = (z + 1)C(z) \rightarrow C(z) = z^3 + (1 - 2i)z^2 + (-1 - 2i)z - 1$$

$$C(-1) = -1 + 1 - 2i + 1 + 2i - 1 = 0 \Rightarrow z_3 = -1$$

$$\begin{array}{c|cccc} & 1 & 1-2i & -1-2i & -1 \\ -1 & \downarrow & -1 & 2i & 1 \\ \hline & 1 & -2i & -1 & 0 \end{array}$$

$$C(z) = (z + 1)E(z) \rightarrow E(z) = z^2 - 2iz - 1 \rightarrow D = (-2i)^2 - 4 \cdot -1 = 0$$

$$z_{4,5} = \frac{2i}{2} = i \Rightarrow E(z) = (z - i)^2$$

$$A(z) = (z + 1) \cdot B(z) \cdot C(z) \cdot E(z) = (z + 1)^3(z - i)^2$$

$$z_{1,2,3,4,5} \in \{-1, -1, -1, i, i\}$$

References